

DANIEL VARTANIAN

I'm a Media Manager at **Talent** Advertising Agency (**Publicis Groupe**) and a Academic Collaborator at the University of São Paulo (**USP**). My work focuses on spatial analysis, **complex systems** modeling, and computational approaches to marketing, social, and public health challenges.

I develop open-source research software, primarily in **R**, **Python**, and **NetLogo**. My tools are used by researchers studying complex systems, **chronobiology**, food security, and climate change.

I'm also involved in building scientific communities and contributing to open-source science through **rOpenSci**.



Access this CV online at:
danielvartan.com/cv

EDUCATION

2025
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2022

MSc, Complex Systems Modeling

University of São Paulo (**USP**)

📍 São Paulo, SP, Brazil

- Founded USP's **Agent-Based Modeling Study Group**.

2022
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2016

BSc, Marketing

University of São Paulo (**USP**)

📍 São Paulo, SP, Brazil

WORK EXPERIENCE

Present
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2026

Media Manager

Talent Advertising Agency

📍 São Paulo, SP, Brazil

- **Lead the creation** of Talent's **OOH** (out-of-home) media intelligence unit, developing data products that replace intuition-based decisions with a systematic approach to media planning.
- Develop geospatial analysis and models, including **Marketing Mix Models**, classification, and optimization, to special projects such as fiber and broadband coverage mapping for **Claro**, one of the largest telecom companies in Latin America.
- Build pipelines that turn client and OOH operator databases (**Eletromídia**, **JCDecaux**, **Kallas**, **NEOOH**) into a single source of truth for the whole planning process.
- Connect the agency with OOH tech and adtech vendors (**AllUnite**, **Taptap Digital**, **bebot Digital**, **Logan**) to evaluate integration and measurement capabilities for client campaigns.

2026
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2025

Research Fellow

Center for Metropolitan Studies (USP)

📍 São Paulo, SP, Brazil

- Led geospatial modeling and large-scale data mining on the **FAPESP**-funded **AcessoSAN** project, investigating healthy food access in favelas and low-income urban communities across Brazil.
- Contributed to the development of the **GeoReDUS** project, an open platform of Brazilian intra-municipal and regional data for public policy planning and monitoring.
- Built pipelines to process and analyze millions of records from **SISVAN**, **Brazilian Revenue Service**, **DATASUS**, **IBGE Census**, and climate datasets like **WorldClim 2.1** and **Mapbiomas**.
- Developed and delivered **training in R programming** and other data related tools to approximately 20 researchers.

CONTACT

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REFERENCES

Vitor Kamada

BETC Havas
Chief Media Officer (CMO)

Profa. Dra. Mariana Giannotti

Center for Metropolitan Studies
Polytechnic School
University of São Paulo

Prof. Dr. Camilo R. Neto

Graduate Program in Complex
Systems Modeling
School of Arts, Sciences and
Humanities
University of São Paulo

Developed with
pagedown and *datadrivencv*
R packages.

Source code available at:
github.com/danielvartan/cv

Last updated on September 7, 2026.

2025
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2024

Research Fellow

Sustentarea (FSP-USP)

📍 São Paulo, SP, Brazil

- Led agent-based modeling on the **CNPq**-funded **Global Syndemic** project investigating climate change effects on child nutrition and health in Brazil.
- Developed **NetLogo** models simulating food production systems and nutritional outcomes in children under 5, using climate projections from **Global Climate Models** and **IBGE** data.
- Created the **logolink** R package to provide analytical infrastructure for **NetLogo** simulations.
- Produced **scientific papers**, **pipelines**, and technical reports related to research in public health and climate science.
- Worked with interdisciplinary teams of nutritionists, biologists, statisticians, and health professionals across multiple universities as a member of the **RESICLIMA Network**.

2024
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2022

Graduate Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Developed nested linear models investigating the relationship between human chronotype and latitude using Brazil's largest chronotype dataset (n=115,166).
- Created the **mctq** R package (published on **CRAN**, peer-reviewed by **rOpenSci**) for processing chronotype questionnaires.
- Built pipelines for statistical modeling and data processing.
- Completed **master's thesis** on chronotype and latitude relationships.

2022
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2019

Assistant Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Developed **actigraphy** monitoring protocols and analytical models for circadian rhythm assessment using non-parametric methods, cosinor analysis, periodograms, and actogram visualizations.
- Created the **actverse** R package providing tools for **actigraphy** data analysis.
- Co-authored the first **Brazilian Actigraphy Consensus** and delivered training workshops to 60+ researchers, including a launch workshop at the 18th Brazilian Sleep Congress.
- Contributed to **USP's** Sleep Genetics project investigating associations between **chronotype** and **clock genes** using **actigraphy** and genetic data from saliva samples across multi-centric cohorts in São Paulo and Baependi, Minas Gerais.
- Worked alongside interdisciplinary teams of doctors, biologists, psychologists, and health professionals across multiple research institutions.

2019
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2016

Undergraduate Research Fellow

University of São Paulo (USP)

📍 São Paulo, SP, Brazil

- Provided statistical modeling and data analysis support for **circadian rhythm** and sleep research projects.
- Served as head of marketing and communications for **USP's** Interdisciplinary Sleep Research Group (**GIPSO**), securing media coverage in major Brazilian outlets like *Folha de São Paulo* and *TV Globo's Fantástico*.
- Co-developed sleep science courses through **EACH-USP's** summer program and the **Coursera** platform.

AFFILIATIONS

RESICLIMA Network

Volunteer Academic Collaborator (2024-Present)

Sustentarea

Volunteer Academic Collaborator (2024-Present)

rOpenSci

Software Developer (2021-Present)

Interdisciplinary Sleep Research Group (**GIPSO**)
Researcher (2016-2022)

LANGUAGES

🐦 **danielvartan**

English

🗣️ Fluent

🗣️ Fluent

🗣️ Fluent

🗣️ Fluent

Portuguese

🗣️ Native

🗣️ Native

🗣️ Native

🗣️ Native

Spanish

🗣️ Intermediate

🗣️ Proficient

🗣️ Intermediate

🗣️ Beginner

French

🗣️ Beginner

🗣️ Beginner

🗣️ Beginner

🗣️ Beginner



SELECTED SOFTWARE PROJECTS

2026	PMTilesBR Creator and Maintainer Geospatial tiled datasets for spatial visualization in Brazil.	GitHub
2026	LogoActions Creator and Maintainer GitHub Actions for the NetLogo framework.	GitHub
2025	logolink Creator and Maintainer R package providing a tidy interface between R and NetLogo. - Published on CRAN (4K+ downloads). - Currently under peer review by rOpenSci.	CRAN
2025	LogoScope Creator and Maintainer VS Code extension providing syntax highlighting for NetLogo. Published on Visual Studio Code Marketplace (780 downloads) and Open VSX (2.2K downloads).	Visual Studio Code Marketplace
2025	LogoClim Creator and Maintainer NetLogo model developed to simulate and visualize climate conditions using data from WorldClim 2.1. - Published on CoMSES Net. - With an accompanying article published in JOSS.	CoMSES Net
2024	brandr Creator and Maintainer R package for managing brand visual identity in data products using the brand.yml standard. Published on CRAN (3K+ downloads).	CRAN
2023	actverse Creator and Maintainer R package providing a tidy toolbox for actigraphy data analysis.	GitHub
2021	mctq Creator and Maintainer R package providing a tidy toolbox for processing the Munich ChronoType Questionnaire (MCTQ). - Published on CRAN (15K+ downloads). - Peer reviewed by rOpenSci. - Cited in chronobiology research across multiple countries.	CRAN

PROGRAMMING LANGUAGES

danielvartan

Expert

R

NetLogo

Fluent

Python

SQL

Intermediate

Bash

C C++ C#

GAML (GAMA)

JavaScript

Lua

Basic Proficiency

GDScript (Godot)

Julia

Rust

Markup/Other Languages

Git

Docker

HTML

CSS

SASS

Markdown

TeX LaTeX



TEACHING EXPERIENCE

2026
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2025

An Introduction to the R Programming Language

University of São Paulo

📍 São Paulo, SP, Brazil

- Developed and delivered an introductory R programming course for researchers at the School of Public Health (**FSP**) (2025) and School of Philosophy, Literature and Human Sciences (**FFLCH**) (2026) at the University of São Paulo (**USP**).
- Trained approximately 50 researchers across both institutions.

2024

Technical High School Program in Administration

São Paulo State Government

📍 São Paulo, SP, Brazil

- Taught mathematics to approximately 100 students across two public high schools in São Paulo State's technical education program.

2023

ACH0042 - Problem Based Learning II

University of São Paulo

📍 São Paulo, SP, Brazil

- Taught science and computational methods to 60 information systems students at **EACH-USP**.



HONORS, AWARDS & FELLOWSHIPS

2026
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2025

Technical Training (Level IV) Research Fellowship

São Paulo Research Foundation (**FAPESP**)

2025
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2024

Technological and Industrial Development (DTI-B) Research Fellowship

National Council for Scientific and Technological Development (**CNPq**)

2024
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2022

Master's Research Fellowship

Coordination for the Improvement of Higher Education Personnel (**CAPES**)



SELECTED OUTREACH PROJECTS

Present
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2026

Interdisciplinary Brazilian Symposium on Agent-Based Models (SIMBRA**)**

University of São Paulo

📍 São Paulo, SP, Brazil

Co-founded SIMBRA as an extension of USP's Agent-Based Modeling Study Group (founder) to provide a national platform for researchers and students across disciplines to discuss and disseminate agent-based modeling approaches.

2018
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2017

Durma na EACH (Sleep at EACH)

University of São Paulo

📍 São Paulo, SP, Brazil

Created sleep and rest spaces for students at **EACH-USP** with plans for expansion to São Paulo public schools.



SELECTED PUBLICATIONS

- 2026 • **LogoClim: WorldClim in NetLogo**
Journal of Open Source Software
Vartanian, D., Garcia, L., & Carvalho, A. M. (2026). LogoClim: WorldClim in NetLogo. *Journal of Open Source Software*, 11(124), 8845. <https://doi.org/10.21105/joss.08845>
- 2026 • **rOpenSci Packages: Development, Maintenance, and Peer Review**
rOpenSci
Equipe editorial da rOpenSci (2026). *Pacotes rOpenSci: Desenvolvimento, manutenção e revisão por pares* [rOpenSci packages: Development, maintenance, and peer review] (Portuguese translation: Pedro Faria, Francesca Palmeira, Beatriz Milz, Samuel Carleial Fernandes, Rafael Fontenelle, Ariana Moura Cabral, Ildeberto Vasconcelos, Daniel Vartanian, Marcelo Perlin, João Granja-Correia, António Pedro) Zenodo. <https://zenodo.org/doi/10.5281/zenodo.18833361>
- 2025 • **Is Latitude Associated with Chronotype?**
University of São Paulo
Vartanian, D. (2025). *Is latitude associated with chronotype?* [Master's thesis, University of São Paulo]. <https://doi.org/10.11606/D.100.2025.tde-02042025-063648>
- 2023 • **Actigraphic Characterization of Sleep and Circadian Phenotypes of PER3 Gene VNTR Genotypes**
Chronobiology International
Viana-Mendes, J., Benedito-Silva, A. A., Andrade, M. A. M., Vartanian, D., Gonçalves, B. da S. B., Cipolla-Neto, J., & Pedrazzoli, M. (2023). Actigraphic characterization of sleep and circadian phenotypes of PER3 gene VNTR genotypes. *Chronobiology International*, 40(9), 1244–1250. <https://doi.org/10.1080/07420528.2023.2256858>
- 2021 • **Brazilian Actigraphy Consensus**
Segmento Farma Editores
Pedrazzoli, M., & Gonçalves, B. S. B. (Eds.) (with Benedito-Silva, A. A., Toscanini, A. C., Souza, A. P. P. B., Conway, B. A., Vartanian, D., Santos, E. H. R., Pires, M. L. N., & Leocadio-Miguel, M. A.). (2021). *Consenso brasileiro de actigrafia* [Brazilian actigraphy consensus]. Segmento Farma Editores. ISBN 978-65-992059-6-5.
- 2020 • **Early Chronotype with Advanced Activity Rhythms and Dim Light Melatonin Onset in a Rural Population**
Journal of Pineal Research
Ruiz, F. S., Bejjamini, F., Beale, A. D., Gonçalves, B. D. S. B., Vartanian, D., Taporoski, T. P., Middleton, B., Krieger, J. E., Vallada, H., Arendt, J., Pereira, A. C., Knutson, K. L., Pedrazzoli, M., & Von Schantz, M. (2020). Early chronotype with advanced activity rhythms and dim light melatonin onset in a rural population. *Journal of Pineal Research*, 69(3). <https://doi.org/10.1111/jipi.12675>